

ML-MDS Lab Project Proposal

Machine Learning Pipeline on Bank Marketing Dataset

Arman Bazarchi

Ines Maria Madeira Prates

Contents

1	Problem Definition	1
2	Dataset Description	1
3	Motivation	1

1 Problem Definition

Our objective for this project is to design and evaluate a machine learning pipeline for a real-world classification problem, specifically, we will address the task of predicting whether a client will subscribe to a term deposit based on information collected during direct marketing campaigns conducted by a banking institution.

In this pipeline we might face challenges such as careful preprocessing and feature selection of a mix of numerical and categorical variables, also our target variable is typically imbalanced, as only a small proportion of clients subscribe to the product, another challenge can be the data including campaign-related features (e.g., number of contacts, previous outcomes) that may introduce complex, non-linear relationships with the target variable.

2 Dataset Description

We use the original version of Bank Marketing dataset from the [UCI Machine Learning Repository](#). It consists of a large number of client records derived from direct telephone marketing campaigns. The dataset includes both numerical and categorical variables and presents challenges such as class imbalance and the presence of potentially redundant or highly correlated features.

The target is binary, indicating whether a client subscribed to a term deposit. Features describe client characteristics and marketing interactions, including demographic (age, job, marital status, education), financial (loans and credit status), and campaign-related variables (contact type, timing, number of contacts, and previous outcomes). Duration of the last contact is highly predictive, but since it is only available after the call, we may consider excluding it or analyzing it separately to avoid data leakage.

The dataset shows a strong imbalance between the two classes.

- **No subscription:** 36,548 samples (88.73%)
- **Yes subscription:** 4,640 samples (11.27%)

Such imbalance is important for model design and evaluation, as it may bias models toward predicting the majority class. Therefore, accuracy alone is not a sufficient evaluation metric. Instead, we may rely on metrics such as F1-score, ROC-AUC, and AUPR, which are more robust under class imbalance conditions.

3 Motivation

This dataset allows us to explore customer behavior, possibility of having non-linear relationships, evaluate different modeling assumptions, and analyze the impact of feature selection and regularization, also class imbalance makes it a realistic scenario for comparing multiple approaches.

From a practical perspective, accurately predicting whether a client will subscribe to a term deposit is highly valuable for financial institutions, as it enables more efficient targeting of marketing campaigns, reduces operational costs, and improves conversion rates.

We plan to apply PCA and clustering (K-Means, GMM) for structural analysis, followed by Logistic Regression with regularization, SVM (linear, polynomial, RBF), and ensemble methods (Random Forest, Gradient Boosting). Models will be evaluated using cross-validation and metrics suitable for imbalanced data.